ARM BASED EMBEDDED SYSTEM DESIGN LAB

CA Activity-1

NAME: Om.G.Sapdhare **DIV:** C

PRN:202502070030 **BATCH:** C4

- **Problem Statement: Design and implement an embedded system that performs the following operations:**

1. Switch 1 pressed

> Turn ON LED

> Display digit 1 on 7-segment display

2. Switch 2 pressed

> Turn ON Buzzer

> Display digit 2 on 7-segment display

3. Switch 3 pressed

> Activate Relay

> Display digit 3 on 7-segment display

4. No switch pressed

> Turn OFF all outputs

- **System Requirements:**

1. Hardware Requirements

- **STM32 Nucleo Board** (e.g., STM32F401RE)
- Push Buttons (3 switches for input)
- LED (for visual indication)
- Buzzer (for sound alert)
- Relay Module (for switching external load)
- 7-Segment Display (Common Cathode preferred)
- Resistors (220Ω for LED, 10kΩ for switches)
- Breadboard and Connecting Wires
- USB Cable (for programming and power supply)

2. Software Requirements

- **STM32CubeIDE** (for coding and debugging)
- **STM32CubeMX** (for GPIO configuration and code generation)
- Embedded C Programming Language
- **Proteus** (optional, for simulation and demonstration)

- **Circuit Diagram Screenshot:**

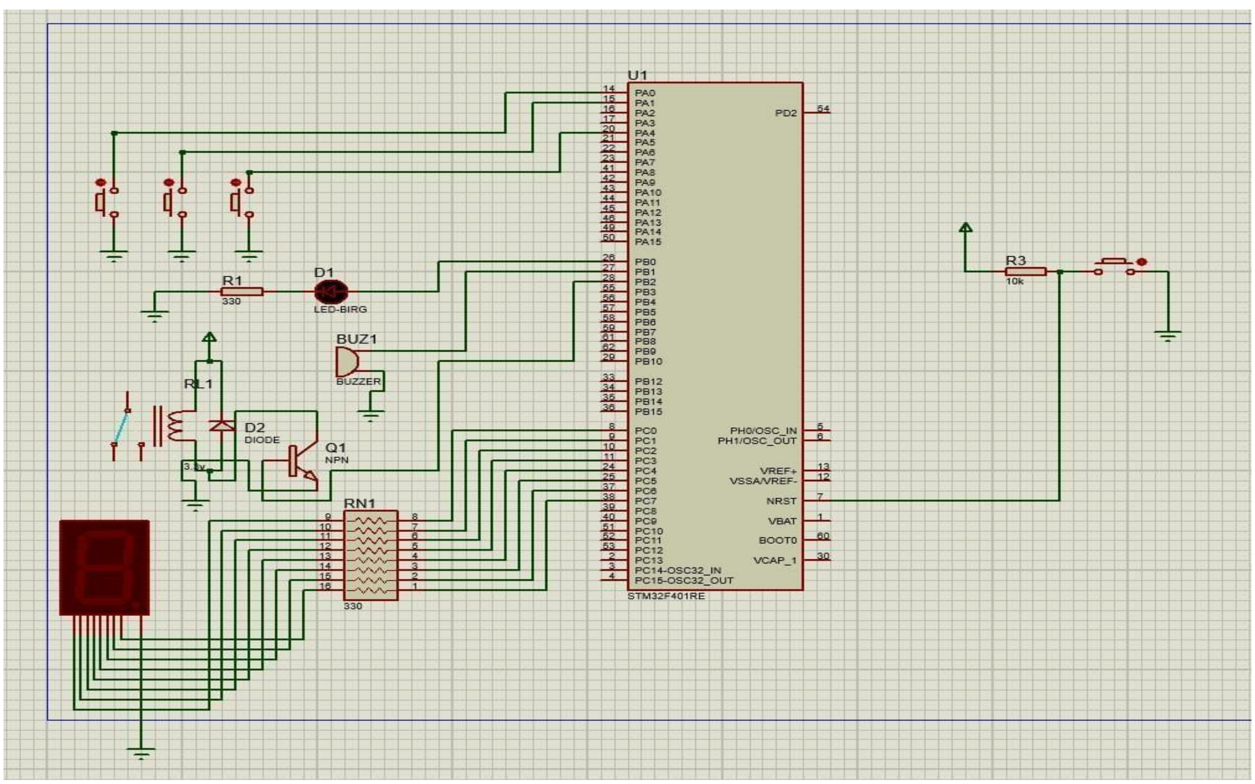


Fig.Implementation of a smart alert system using STM32.

- **STM32 Embedded C Program Using HAL Drivers:**

  GPIO Configuration
- Input (Pull-down)
- Output (Push-pull)

  HAL Functions Used
- HAL_GPIO_ReadPin() → To read switch input
- HAL_GPIO_WritePin() → To control LED, buzzer, relay
- HAL_Delay() → For switch debouncing

  HAL_GPIO_ReadPin() is used to detect whether a switch is pressed or not by reading the logic level of the GPIO pin.

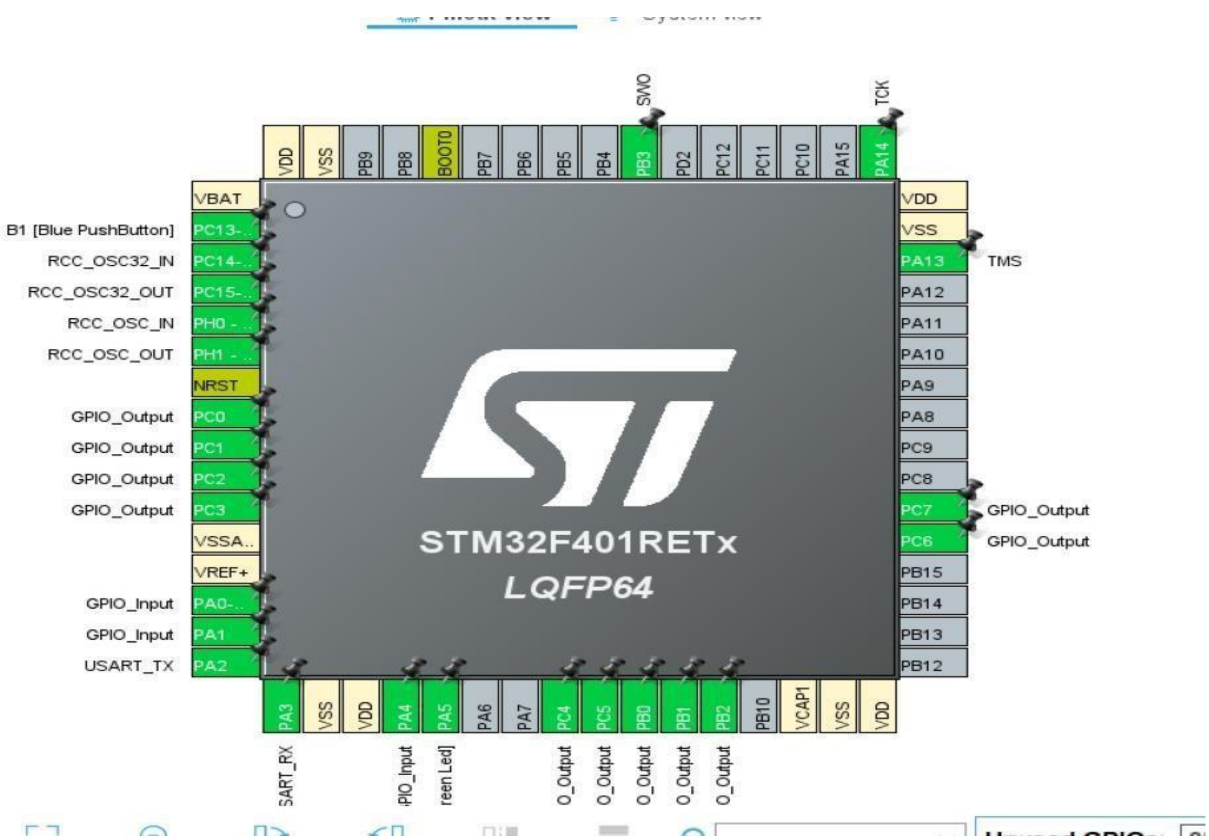


Fig. Configure GPIO in Cube MX

- **HAL-Based Code:**

```
while (1)

{

// --- CONDITION 1: Switch 1 Pressed (PA0) ---

if (HAL_GPIO_ReadPin(GPIOA, GPIO_PIN_0) == GPIO_PIN_RESET)

{

// LED ON (PB0) | Buzzer & Relay OFF (PB1, PB2)

HAL_GPIO_WritePin(GPIOB, GPIO_PIN_0, GPIO_PIN_SET);

HAL_GPIO_WritePin(GPIOB, GPIO_PIN_1 | GPIO_PIN_2, GPIO_PIN_RESET);


// Digit 1: Segments b, c ON | a, d, e, f, g OFF

HAL_GPIO_WritePin(GPIOC, GPIO_PIN_1 | GPIO_PIN_2, GPIO_PIN_SET);

HAL_GPIO_WritePin(GPIOC, GPIO_PIN_0 | GPIO_PIN_3 | GPIO_PIN_4 | GPIO_PIN_5 |
GPIO_PIN_6, GPIO_PIN_RESET);

}

// --- CONDITION 2: Switch 2 Pressed (PA1) ---

else if (HAL_GPIO_ReadPin(GPIOA, GPIO_PIN_1) == GPIO_PIN_RESET)

{

// Buzzer ON (PB1) | LED & Relay OFF (PB0, PB2)

HAL_GPIO_WritePin(GPIOB, GPIO_PIN_1, GPIO_PIN_SET);

HAL_GPIO_WritePin(GPIOB, GPIO_PIN_0 | GPIO_PIN_2, GPIO_PIN_RESET);

// Digit 2: Segments a, b, d, e, g ON | c, f OFF

HAL_GPIO_WritePin(GPIOC, GPIO_PIN_0 | GPIO_PIN_1 | GPIO_PIN_3 | GPIO_PIN_4 |
GPIO_PIN_6, GPIO_PIN_SET);

        HAL_GPIO_WritePin(GPIOC, GPIO_PIN_2 | GPIO_PIN_5, GPIO_PIN_RESET);
```

```
    }

    // --- CONDITION 3: Switch 3 Pressed (PA4) ---
    else if (HAL_GPIO_ReadPin(GPIOA, GPIO_PIN_4) == GPIO_PIN_RESET)
    {
      // Relay ON (PB2) | LED & Buzzer OFF (PB0, PB1)
      HAL_GPIO_WritePin(GPIOB, GPIO_PIN_2, GPIO_PIN_SET);
      HAL_GPIO_WritePin(GPIOB, GPIO_PIN_0 | GPIO_PIN_1, GPIO_PIN_RESET);

      // Digit 3: Segments a, b, c, d, g ON | e, f OFF
      HAL_GPIO_WritePin(GPIOC, GPIO_PIN_0 | GPIO_PIN_1 | GPIO_PIN_2 |
GPIO_PIN_3 | GPIO_PIN_6, GPIO_PIN_SET);
      HAL_GPIO_WritePin(GPIOC, GPIO_PIN_4 | GPIO_PIN_5, GPIO_PIN_RESET);
    }

    // --- CONDITION 4: No Switch Pressed ---
    else
    {
      // All Actuators OFF (PB0, PB1, PB2)
      HAL_GPIO_WritePin(GPIOB, GPIO_PIN_0 | GPIO_PIN_1 | GPIO_PIN_2,
GPIO_PIN_RESET);

      // All Segments OFF (PC0 to PC6)
      HAL_GPIO_WritePin(GPIOC, GPIO_PIN_0 | GPIO_PIN_1 | GPIO_PIN_2 |
GPIO_PIN_3 | GPIO_PIN_4 | GPIO_PIN_5 | GPIO_PIN_6, GPIO_PIN_RESET);
    }
  }
```

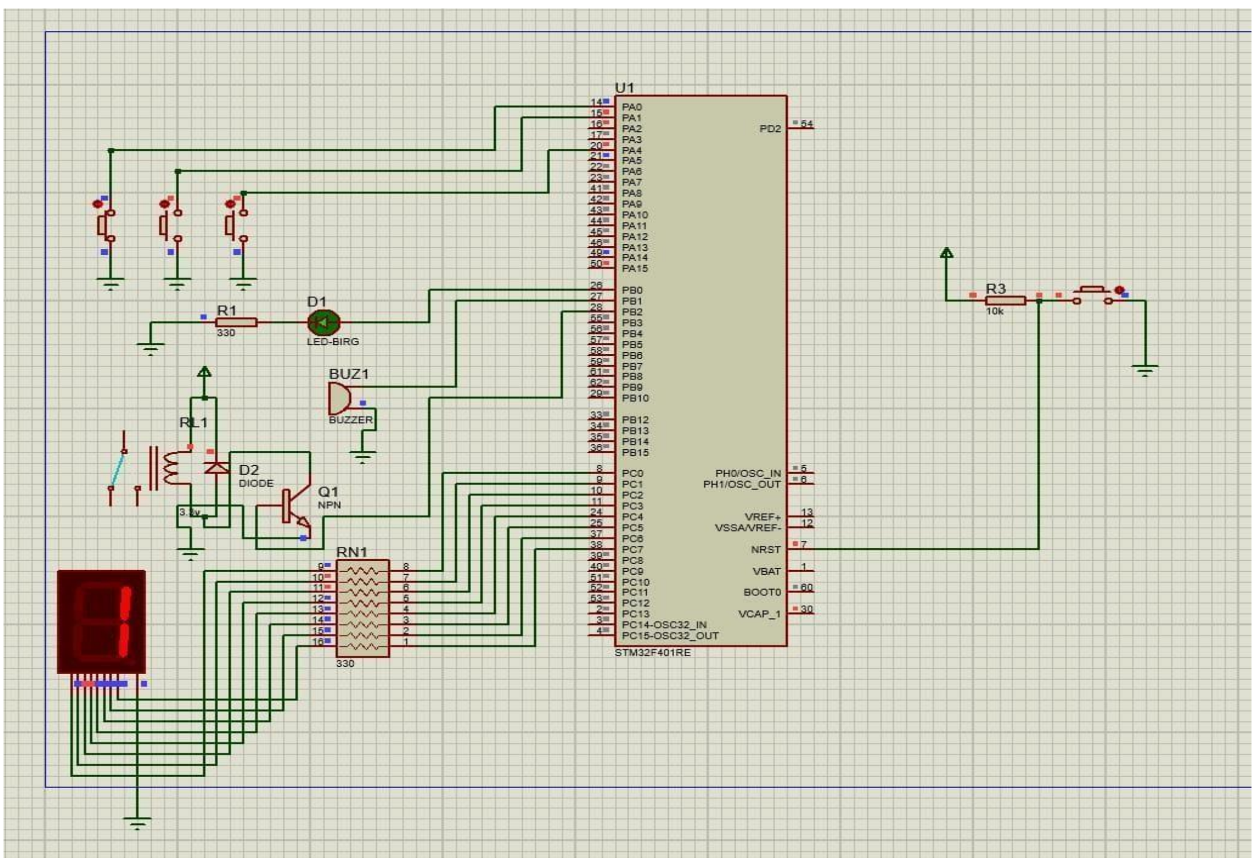
- **Prototype Working & Program logic on Proteus:**

Program Logic-

1. Initialize system using HAL_Init()
2. Configure GPIO pins using cube MX
3. Continuously check switch status
4. Based on switch:
   - Turn ON corresponding output
   - Display digit on 7-segment
5. If no switch → Turn OFF all outputs

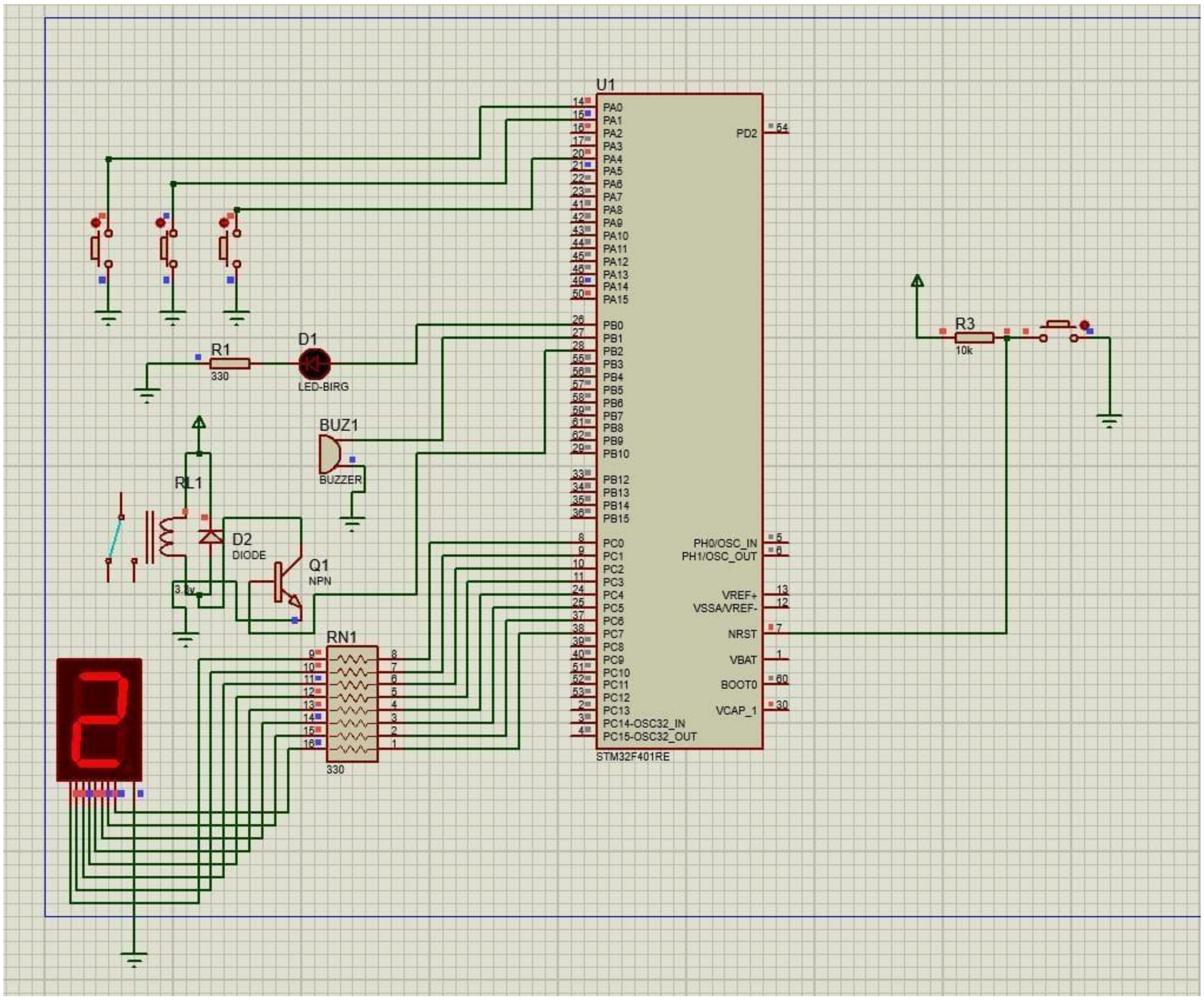
1. When Switch 1 is Pressed-

Switch 1 → LED ON and display “1”



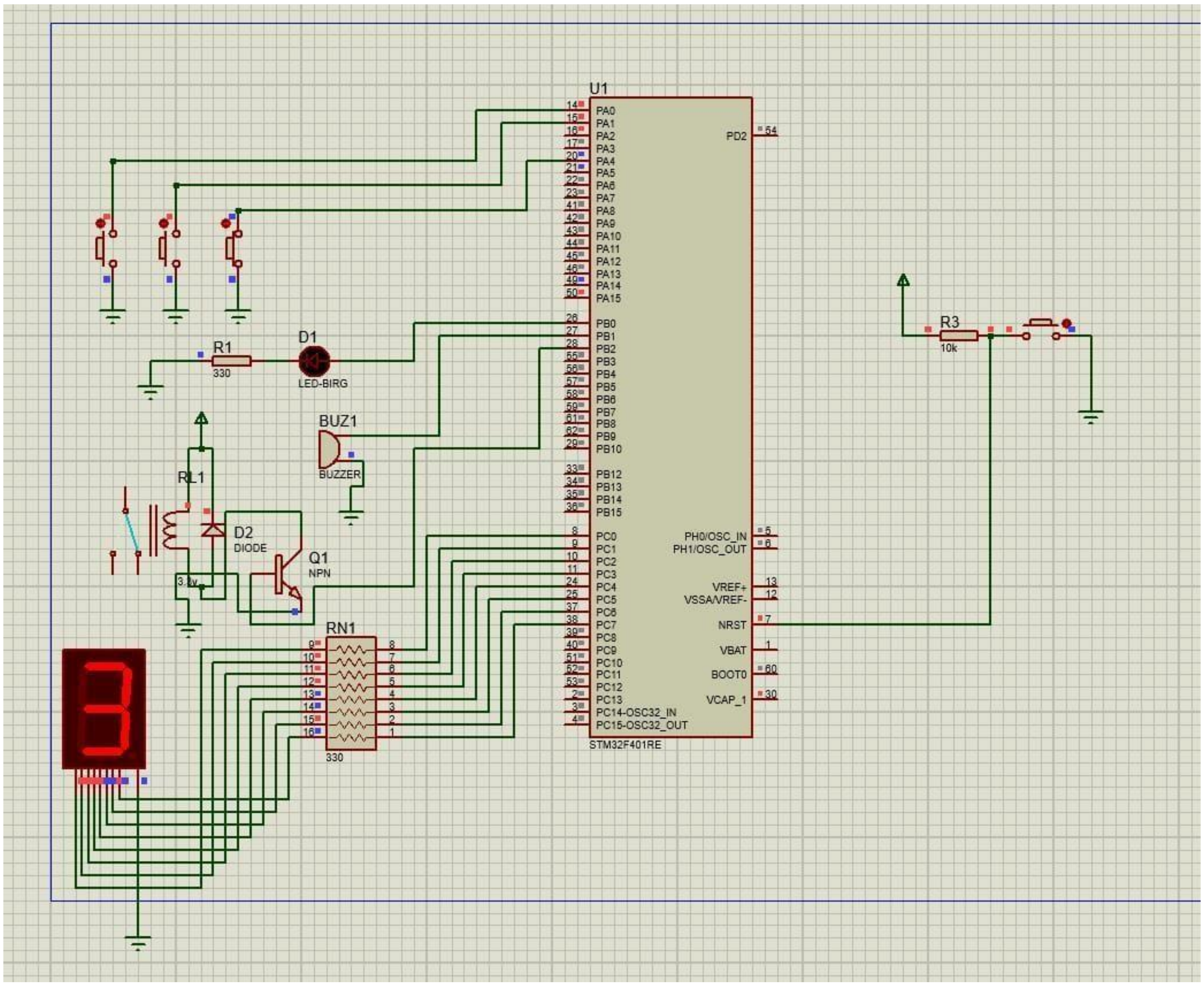
2. When Switch 2 is Pressed-

Switch 2 → Buzzer ON and display “2”



3. When Switch 3 is Pressed:

Switch 3 → Relay ON and display “3”



4. No Switch is Pressed:

“Turn OFF all Outputs”